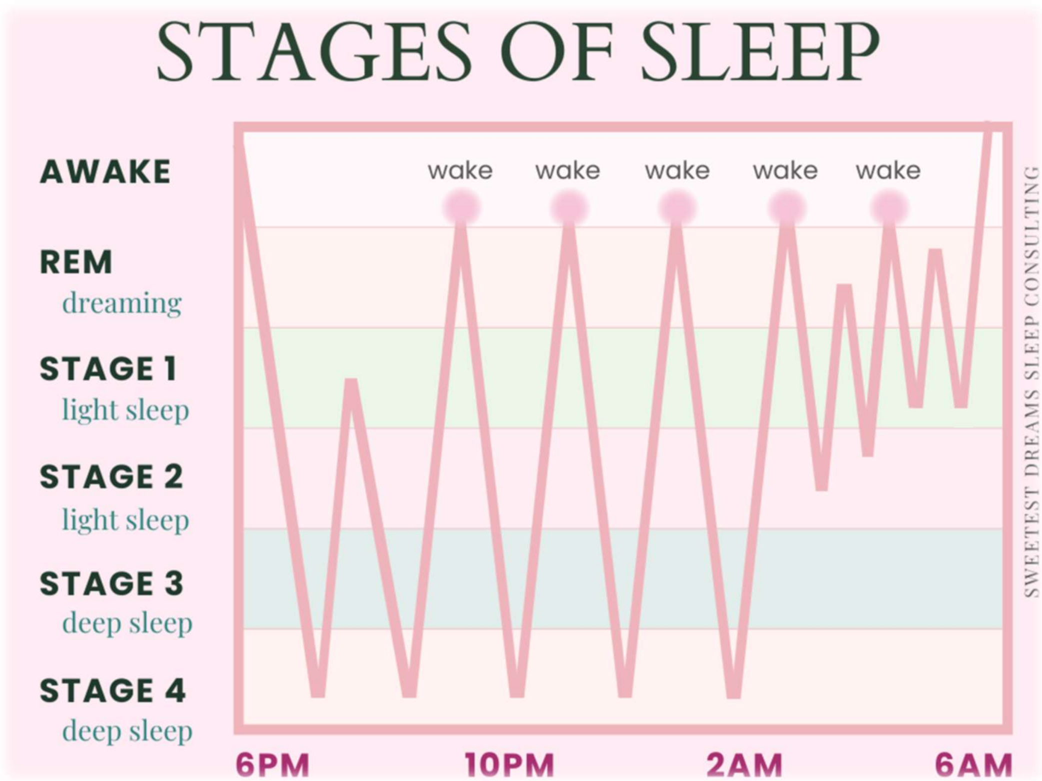
Biomedical Department

Signal and Systems

Sleep Stages Classification using EOG Signals



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➢ ***Introduction:***

Sleep Stage Classification

Sleep is a vital process for human health and well-being. Changes in sleep patterns are closely linked to serious health issues like heart disease, diabetes, obesity, and neurological disorders. Identifying sleep stages accurately is crucial for diagnosing and monitoring sleep problems. The standard method for classifying sleep stages is Polysomnography (PSG). This technique records multiple physiological signals at the same time, including Electroencephalography (EEG), Electrooculography (EOG), Electromyography (EMG), Electrocardiography (ECG), and breathing signals. According to the American Academy of Sleep Medicine (AASM) guidelines, sleep is divided into five stages: Wakefulness (W), Non-REM Stage 1 (N1), Non-REM Stage 2 (N2), Non-REM Stage 3 (N3), and Rapid Eye Movement (REM) sleep. Each stage is scored in 30-second intervals.

Why EOG?

Among the signals recorded during PSG, the Electrooculogram (EOG) captures the electrical activity from eye movements. EOG is particularly useful for sleep staging for several reasons. First, it is the best signal for identifying Rapid Eye Movement (REM) sleep since REM is marked by rapid, conjugated eye movements that create clear and recognizable patterns in the EOG signal. Second, EOG shows significant physiological changes across all sleep stages: during wakefulness, eye movements are frequent and erratic; during N1, slow rolling eye movements occur and during N2 and N3, movements are minimal or absent. Third, EOG electrodes are easier to place and less affected by scalp impedance issues compared to EEG electrodes, which makes EOG more practical for both clinical and home monitoring.

Why Not Other Signals?

While EEG is considered the best standard for sleep staging due to its detailed view of brain activity, it needs many scalp electrodes and is very sensitive to electrode placement and impedance. This makes it less practical outside clinical environments. ECG can provide valuable heart rate information, but it does not have the resolution to differentiate between similar stages like N1 and N2. EMG mainly detects muscle relaxation during REM sleep and gives limited insight into other sleep stages on its own. Breathing signals such as airflow and chest effort are more relevant for breathing disorders like sleep apnea than for classifying sleep stages. In contrast, EOG balances detailed information

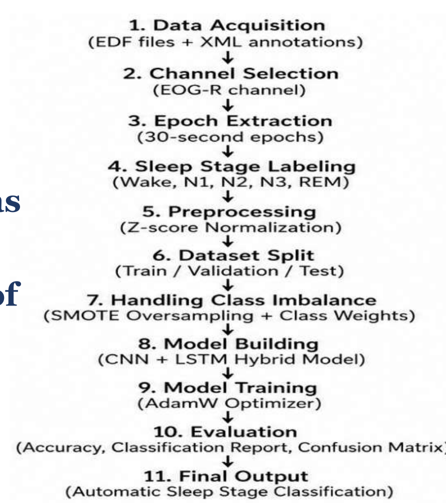
with ease of use, offering specific eye movement patterns that are essential for automated classification, especially in distinguishing REM from non-REM sleep.

MESA Dataset From NSRR

The data for this study came from the Multi-Ethnic Study of Atherosclerosis (MESA) Sleep Study, which is publicly available through the National Sleep Research Resource (NSRR). MESA is a large, long-term study that began in 2000 to investigate the prevalence and progression of early cardiovascular disease in six major metropolitan areas in the United States. The study included 6,814 participants from diverse ethnic backgrounds, including White, African American, Hispanic, and Chinese communities, who were aged 45 to 84 at the start. The sleep aspect of MESA, called the MESA Sleep Exam, was introduced to look at the connection between sleep patterns and cardiovascular outcomes. Each participant had a full overnight PSG recording done at home with standardized equipment. The PSG recordings are saved in European Data Format (EDF) files, each containing multiple simultaneously recorded physiological channels, including EEG, EOG, EMG, ECG, airflow, thoracic and abdominal breathing effort, oxygen saturation, heart rate, and body position, all sampled at 256 Hz. Each EDF file is accompanied by an Extensible Markup Language (XML) annotation file that follows the NSRR standard format. This file includes the analysis of the signals made by Specialized doctors in which each epoch in the signal is labeled by a certain event.

For this Project, ten PSG recordings were chosen from the MESA Sleep dataset. The EOG-R channel was extracted from each recording as the main input signal. A total of 12,758 intervals were obtained across all recordings, offering a full and representative sample for training and evaluating the proposed deep learning model.

➢ *Workflow and Code:*



Signal Exploration

Before loading the data, a single PSG recording was explored to examine the general properties of the dataset. This step aimed to verify the consistency of the recording parameters across subjects and to understand the signal characteristics before processing the entire dataset. The first patient recording was loaded using the MNE library with the preload=False parameter. This setting reads only the file metadata without loading the raw signal data into memory, making it an efficient choice as it avoids unnecessary memory use during exploration. The exploration showed that each recording lasts about 12 hours of overnight sleep, sampled at 256 Hz across 27 physiological channels.

Channel Display

After the initial exploration, all available channels were listed within the EDF recording to identify the signals for analysis and justify our channel selection for the next step. The 27 recorded channels include neurological, ocular, muscular, cardiac, and respiratory signals. The complete list includes EEG1, EEG2, EEG3, EOG-L, EOG-R, EMG, EKG, airflow, thoracic effort, abdominal effort, oxygen saturation, heart rate, snoring, body position, and several offset channels. According to the AASM guidelines and the goals of this study, the EOG-R channel was chosen as the main input signal for the classification model, as explained in the next section.

Data Loading & Channel Selection

The complete dataset was loaded by iterating over all available EDF files within the designated directory. The glob library was used to automatically detect all files with the .edf extension, eliminating the need to manually specify individual file paths. The EOG-R channel was chosen as the only input signal for this study. As mentioned in the Introduction, EOG provides strong information for distinguishing sleep stages, particularly for identifying REM sleep through its characteristic rapid eye movements. We preferred the right EOG channel (EOG-R) over the left (EOG-L) based on a comparison done during model development, where EOG-R consistently showed higher training stability and overall accuracy.

Sleep Stages Mapping

The sleep stage labels in the XML annotation files are stored as descriptive string identifiers following the NSRR harmonized scoring format. To enable numerical processing by the deep learning model, a mapping dictionary was defined to convert each stage label into an integer class index. The mapping assigns integer labels from 0 to 4 for the five AASM sleep stages: Wakefulness (W), N1, N2, N3, and REM respectively.

Reading EDF & XML Files

For every subject, both the EDF signal file and its corresponding XML annotation file were loaded. The EDF file was read using the MNE library with preload=True, which loads the complete signal data into memory for efficient epoch extraction later.

The XML file was parsed using the xmltodict library, converting the hierarchical XML structure into a Python dictionary. The scored events were extracted from the PSGAnnotation/ScoredEvents/ScoredEvent path, which contains a list of all annotated events recorded during the overnight study. The scored events in the XML file include various annotations beyond sleep stages, such as respiratory events, episodes of oxygen desaturation, artifact markers, and technical recording events. Since we only needed sleep stage labels for the classification task, the events were filtered to keep only the relevant ones.

Events Filtering

Each event in the XML file includes an EventConcept field that describes the event type. Sleep stage events are identified by the keywords "sleep" or "wake", regardless of letter case. For each kept event, three attributes were extracted: the stage label, the start time in seconds from the beginning of the recording, and the duration in seconds. These attributes help locate and extract the corresponding signal segments from the EDF file.

Segmentation

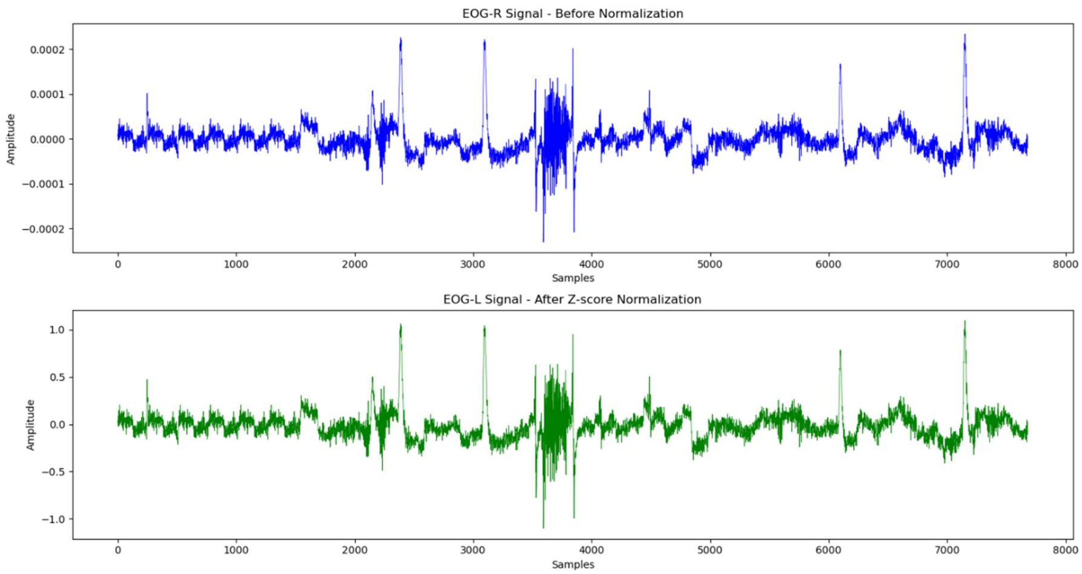
The segmentation step changes the continuous PSG recording into discrete, fixed-length epochs suitable for model input. Although the XML annotation file groups consecutive epochs of the same sleep stage into a single event with a cumulative duration, AASM standards require each 30-second interval to be treated as an independent epoch with its own label.

For each annotated event, the total duration was divided into 30-second windows, and the start time of each window was calculated. The corresponding EOG-R signal segment was then extracted from the EDF file by converting the time offset into a sample index using the known sampling rate. Following segmentation, the dataset consists of 12,758 epochs, each represented as a one-dimensional array of 7,680 amplitude values with its corresponding sleep stage label.

Normalization

Raw EOG signals show amplitude variations both within and across subjects because of differences in electrode placement, skin resistance, and individual physiological traits. To handle this variability and ensure consistent input scaling for the neural network, Z-score normalization was applied to the entire dataset.

Z-score normalization was preferred over Min-Max scaling because EOG signals often contain brief high-amplitude spikes from blinks or artifacts, and Min-Max scaling is very sensitive to these outliers as extreme values compress the remaining signal into a narrow range. Z-score normalization is more robust against such outliers, preserving the signal's relative structure.



Data Splitting

Following normalization, the dataset was partitioned into three subsets: training, validation, and testing, using a stratified random splitting strategy with a ratio of 80%, 10%, and 10% respectively.

Stratified splitting was applied using the stratify parameter to ensure that the proportional distribution of each sleep stage class is preserved across all three subsets. Without stratification, random splitting may result in subsets with disproportionate class distributions, leading to biased model evaluation.

Following splitting, the data arrays were reshaped from the format (epochs, channels, timesteps) to (epochs, timesteps, channels) to conform to the input format expected by the one-dimensional convolutional layers of the proposed model.

Random Splitting VS. Patient Based

In the present study, epoch-level random splitting was adopted because the dataset comprises only ten PSG recordings. Under a patient-based splitting scheme with an 80/10/10 ratio, the validation and testing sets would each contain a single subject. The sleep patterns, stage distributions, and EOG signal characteristics of a single individual are highly subject-specific and may not reflect the diversity of the full dataset, resulting in evaluation metrics that are strongly influenced by the characteristics of that subject rather than the model's true generalization ability. This was confirmed during preliminary experiments, in which patient-based splitting produced a validation accuracy of approximately 17%, compared to 80% under random splitting, indicating that the single validation subject was substantially different from the training population rather than reflecting genuine model failure.

Handling Class Imbalance

SMOTE

Sleep stage data is inherently imbalanced, as the distribution of sleep stages across a typical night does not follow a uniform distribution. In the present dataset, the Wake and N2 stages are considerably overrepresented, while N1 and N3 are substantially underrepresented. This imbalance poses a significant challenge for model training, as the model tends to favor the majority classes and exhibit poor performance in minority classes.

To address this issue, the Synthetic Minority Oversampling Technique (SMOTE) was applied exclusively to the training set. SMOTE generates synthetic samples for minority classes. For each minority class sample, SMOTE selects one of its k nearest neighbors and creates a new synthetic sample along the line segment connecting the two points in the feature space.

SMOTE was intentionally applied only to the training set and not to the validation or testing sets. Applying oversampling to the validation or testing sets would introduce artificial samples that do not reflect the true data distribution, resulting in misleading evaluation metrics. The validation and testing sets were therefore preserved in their original form to provide an unbiased assessment of model generalization

Since SMOTE operates on two-dimensional arrays, the training data was temporarily reshaped from (epochs, timesteps, channels) to (epochs,

features) before applying the oversampling and subsequently restored to its original three-dimensional shape

Before SMOTE VS. After SMOTE

```
✅ Train: (10206, 7680, 1)
✅ Val:   (1276, 7680, 1)
✅ Test:  (1276, 7680, 1)

Classes distribution before SMOTE:
  Wake: 4455
  N1: 754
  N2: 3293
  N3: 695
  REM: 1009
```

```
✅ Train: (22275, 7680, 1)
✅ Val:   (1276, 7680, 1)
✅ Test:  (1276, 7680, 1)

Classes distribution after SMOTE:
  Wake: 4455
  N1: 4455
  N2: 4455
  N3: 4455
  REM: 4455
```

Class Weights

In addition to SMOTE, class weights were incorporated into the model training process as a complementary strategy for handling class imbalance. While SMOTE addresses imbalance by augmenting training data, class weights modify the loss function to assign higher penalties to misclassifications of minority classes, effectively increasing their influence on the weight update process during training.

Model Architecture

The proposed model is a deep learning architecture combining one-dimensional Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks, designed to capture both local features and temporal dependencies within the EOG signal.

Conv1D(32, kernel_size=50, activation='relu')

The first convolutional layer applies 32 learnable filters, each of length 50 samples, to the input EOG signal. Each filter slides along the temporal dimension of the signal, computing the dot product between the filter weights and the corresponding signal segment at each position. This operation enables the layer to detect local patterns and morphological features within windows of approximately 0.2 seconds (50 samples at 256 Hz

MaxPooling1D(pool_size=8)

The max pooling layer reduces the temporal resolution of the feature maps by a factor of 8, retaining only the maximum value within each window of 8 consecutive positions. This operation reduces the computational complexity of subsequent layers by decreasing the length of the feature representations.

Dropout(0.3)

A dropout layer with a rate of 0.3 is applied after each major processing block. During training, dropout randomly deactivates 30% of the neurons in the preceding layer at each forward pass, forcing the learning from different neurons to avoid overfitting.

Conv1D(64, kernel_size=25, activation='relu')

The second convolutional layer applies 64 filters of length 25 samples to the output of the first pooling layer. By increasing the number of filters and reducing the kernel size, this layer is designed to capture more complex and abstract features at a finer temporal resolution. The

hierarchical combination of the two convolutional layers allows the model to learn features at multiple scales, from broad signal trends captured by the first layer to finer patterns detected by the second.

LSTM(32, return_sequences=False)

An LSTM layer with 32 hidden units is used after the convolutional blocks to learn temporal patterns from the extracted features. The LSTM can remember important information over time using gating mechanisms, which helps the model capture long-term dependencies within the EEG epoch. The parameter return_sequences=False means that only the final output of the sequence is passed to the next layer,

Dense(32, activation='relu')

A fully connected dense layer with 32 neurons and ReLU activation is applied after the LSTM layer. This layer combines the temporal features learned by the LSTM and extracts more discriminative patterns useful for sleep stage classification.

Dense(5, activation='softmax')

The final output layer consists of 5 units corresponding to the five sleep stage classes. The softmax activation function normalizes the output into a probability distribution over the five classes. The predicted class is determined by selecting the index with the highest probability.

Model Compilation

The model was compiled using the AdamW optimizer, an extension of the Adam optimizer that incorporates decoupled weight decay regularization.

AdamW was selected over the standard Adam optimizer due to its superior regularization properties. While Adam applies L2 regularization by adding the weight penalty directly to the gradient, AdamW decouples the weight decay from the gradient update, resulting in more effective regularization and improved generalization performance.

A learning rate of 0.0001 was selected to ensure stable and gradual convergence. A larger learning rate risks overshooting the optimal solution, while an excessively small value significantly increases training time. The selected value represents a balance between speed and training stability and was found to effectively reduce ripples in the validation accuracy curve.

A weight decay coefficient of 0.001 was applied to penalize large weight magnitudes during training. This discourages the model from assigning disproportionate importance to any single feature, promoting more distributed and generalizable representations.

Model Training

The model was trained for 80 epochs with a batch size of 64. The batch size of 64 was selected as it provides a balance between speed and accuracy. Smaller batch sizes introduce higher variance in the gradient estimates, leading to increased oscillation in the training curves, while larger batch sizes may reduce the model's ability to generalize.

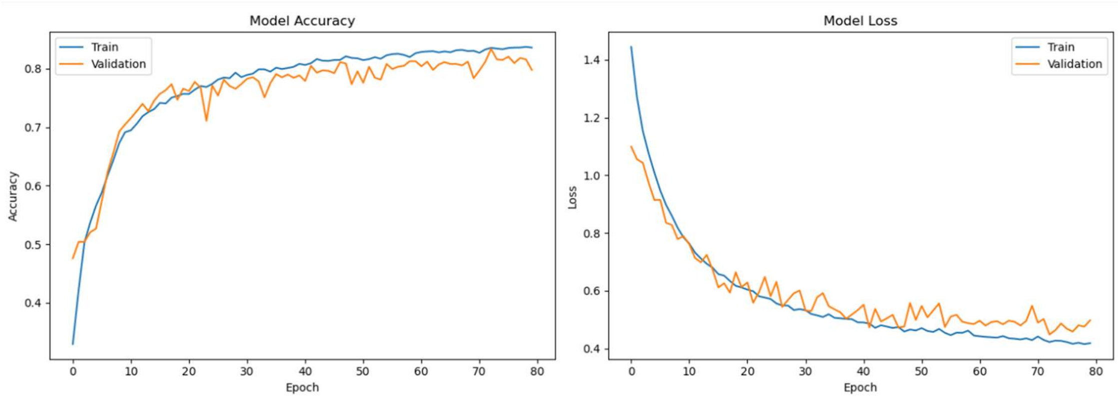
➢ *Results:*

Training Performance

The training and validation accuracy and loss curves over 80 epochs are presented in the figures below. The model demonstrated consistent improvement throughout training, with both training and validation accuracy converging to approximately 80% by the final epoch. The close alignment between the training and validation curves indicates the absence of significant overfitting, suggesting that the regularization strategies employed, including Dropout, AdamW weight decay, and class weighting, were effective in promoting model generalization.

The validation loss exhibited minor oscillations throughout training, which is attributable to the relatively small size of the validation set. Despite these fluctuations, the overall trend of the validation loss

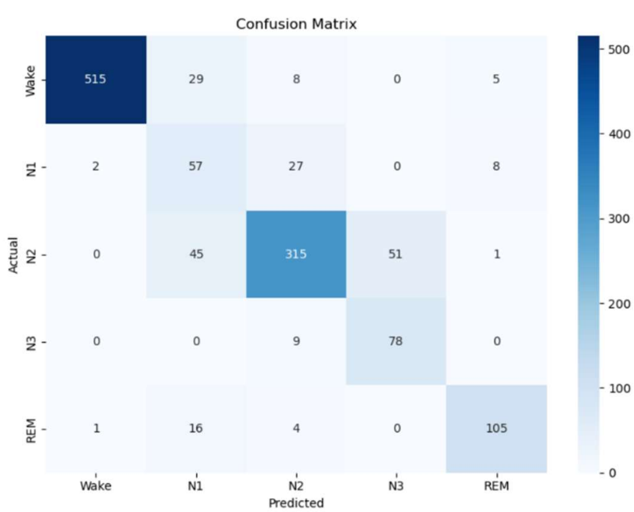
remained downward, confirming that the model continued to improve its generalization ability over the course of training.



Regarding the potential influence of data leakage introduced by epoch-level random splitting, the training dynamics provide important evidence that leakage did not substantially inflate the reported results. If the model had learned to exploit subject-specific patterns present in both the training and testing sets, one would expect the training accuracy to rise rapidly to near-perfect levels while the validation accuracy closely follows due to shared subject information. Instead, both curves exhibit gradual and parallel improvement over 80 epochs, with neither curve reaching artificially high values. The validation accuracy stabilizes around 80% rather than rising to the near-perfect values that would be expected under significant leakage exploitation.

Classification Performance & Analysis of the Results

The performance of the best saved model was evaluated on the held-out test set using the classification report and confusion matrix. The overall test accuracy achieved was 84%



Classification Report:

	precision	recall	f1-score	support
Wake	0.99	0.92	0.96	557
N1	0.39	0.61	0.47	94
N2	0.87	0.76	0.81	412
N3	0.60	0.90	0.72	87
REM	0.88	0.83	0.86	126
accuracy			0.84	1276
macro avg	0.75	0.81	0.76	1276
weighted avg	0.87	0.84	0.85	1276

<u>Wake:</u> Achieved the highest classification performance with an F1-score of 0.95, reflecting the highly distinctive eye movement patterns associated with wakefulness, which are easily separable from sleep stages in the EOG signal.

<u>N2</u>: Demonstrated strong performance with an F1-score of 0.82, consistent with the relatively high frequency of N2 epochs in the dataset and the presence of characteristic EOG patterns during this stage.

<u>N3:</u> Achieved an F1-score of 0.75, which is notable given the limited number of N3 epochs available for training. The application of SMOTE and class weighting contributed to the model's ability to learn meaningful representations for this underrepresented stage.

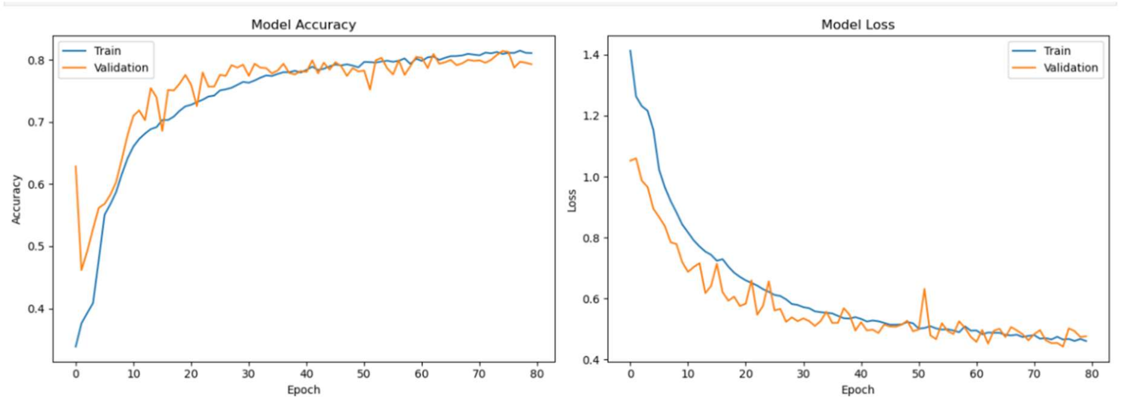
<u>REM:</u> Achieved an F1-score of 0.72. While EOG is theoretically the most informative signal for REM detection, the relatively modest recall of 0.62 suggests that some REM epochs were misclassified as other stages. This may be attributed to the variability in eye movement density across REM periods and the limited number of REM epochs in the dataset.

<u>N1:</u> Exhibited the lowest F1-score of 0.46, with a precision of only 0.33. This is consistent with findings reported in the sleep staging literature, as N1 is a transitional stage with subtle and highly variable EOG characteristics that overlap significantly with both Wake and N2. The low precision indicates that a considerable number of non-N1 epochs were incorrectly predicted as N1, which is a common challenge in automated sleep staging systems.

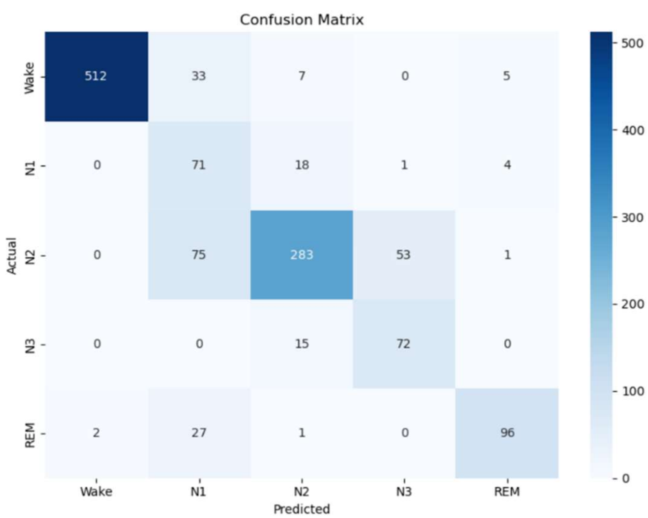
<u>EOG-R VS. EOG-L</u>

A comparative experiment was conducted to evaluate the performance of the model trained on the right EOG channel (EOG-R) versus the left EOG channel (EOG-L) using identical model architecture, hyperparameters, and training procedures. The results demonstrated that EOG-R consistently outperformed EOG-L in terms of overall accuracy, training stability, and per-class F1-scores for Wake, N2, N3, and REM. EOG-L showed marginally better recall for N1. These findings suggest that the right EOG channel provides more discriminative information for sleep stage classification in the MESA dataset, and therefore EOG-R was adopted as the primary input signal for the final model.

Training and validation accuracy and loss curves for EOG-L:



Confusion Matrix and Classification Report for EOG-L:




```
Classification Report:
              precision    recall  f1-score   support

        Wake       1.00      0.92      0.96       557
          N1       0.34      0.76      0.47        94
          N2       0.87      0.69      0.77       412
          N3       0.57      0.83      0.68        87
         REM       0.91      0.76      0.83       126

    accuracy                           0.81      1276
   macro avg       0.74      0.79      0.74      1276
weighted avg       0.87      0.81      0.83      1276
```